

2520090081 - M. SAI RAM PRAGNEY

```
MySQL 8.0 Command Line Cli x + v
-> );
ERROR 1046 (3D000): No database selected
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| bank_db  |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.00 sec)

mysql> USE database_name;
ERROR 1049 (42000): Unknown database 'database_name'
mysql> USE college;
ERROR 1049 (42000): Unknown database 'college'
mysql> CREATE DATABASE college;
Query OK, 1 row affected (0.01 sec)

mysql> USE college;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| college    |
+-----+
1 row in set (0.00 sec)

mysql>
```

Create the table

```
MySQL 8.0 Command Line Cli x + v - □ x

| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> USE database_name;
ERROR 1049 (42000): Unknown database 'database_name'
mysql> USE college;
ERROR 1049 (42000): Unknown database 'college'
mysql> CREATE DATABASE college;
Query OK, 1 row affected (0.01 sec)

mysql> USE college;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| college |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE student_marks (
->     roll_no INT PRIMARY KEY,
->     name VARCHAR(50),
->     subject VARCHAR(50),
->     marks DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> |
```

Insert the data

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| college    |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE student_marks (
->     roll_no INT PRIMARY KEY,
->     name VARCHAR(50),
->     subject VARCHAR(50),
->     marks DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO student_marks
-> VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
```

Display the table

```
MySQL 8.0 Command Line Cli × + - □ ×

-> VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 78.40 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

Aggregate Functions

```
mysql> SELECT COUNT(*) FROM student_marks;
```

COUNT(*)
10

```
1 row in set (0.00 sec)
```

```
mysql> SELECT SUM(marks) FROM student_marks;
```

SUM(marks)
899.60

```
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(marks) FROM student_marks;
```

AVG(marks)
89.960000

```
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(marks) FROM student_marks;
```

MAX(marks)
99.90

```
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

+-----+
| SUM(marks) |
+-----+
|      899.60 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(marks) FROM student_marks;
+-----+
| AVG(marks) |
+-----+
|  89.960000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MAX(marks) FROM student_marks;
+-----+
| MAX(marks) |
+-----+
|      99.90 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MIN(marks) FROM student_marks;
+-----+
| MIN(marks) |
+-----+
|      78.40 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

WHERE

```
mysql> SELECT * FROM student_marks WHERE marks > 85;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

```
7 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks WHERE marks >= 90;
```

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40



```
mysql> SELECT * FROM student_marks WHERE marks >= 90;
```

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
5	Vijay	Math	80.25
10	seetha	azure	82.25

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks WHERE name LIKE 'P%';
```

roll_no	name	subject	marks
4	Priya	Math	88.90

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

mysql> SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 4       | Priya | Math    | 88.90 |
| 5       | Vijay | Math    | 80.25 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks WHERE name LIKE 'P%';
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 4       | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks WHERE name IN ('Ravi','Sita','Vijay');
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 2       | Sita  | Math    | 92.75 |
| 5       | Vijay | Math    | 80.25 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks
-> WHERE marks > 85
-> AND (subject='Math' OR name LIKE 'P%');
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 2       | Sita  | Math    | 92.75 |
| 4       | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

UPDATE

```
MySQL 8.0 Command Line Cli × + ▾
mysql> UPDATE student_marks
-> SET marks = 90
-> WHERE roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi   | Math    | 85.50 |
| 2       | Sita   | Math    | 92.75 |
| 3       | Anil   | Math    | 90.00 |
| 4       | Priya  | Math    | 88.90 |
| 5       | Vijay  | Math    | 80.25 |
| 6       | subbusir | aws     | 98.50 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 9       | kavithamam | aws1   | 99.90 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> UPDATE student_marks
-> SET subject='Remedial Math'
-> WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi   | Math    | 85.50 |
| 2       | Sita   | Math    | 92.75 |
| 3       | Anil   | Math    | 90.00 |
| 4       | Priya  | Math    | 88.90 |
| 5       | Vijay  | Math    | 80.25 |
| 6       | subbusir | aws     | 98.50 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 9       | kavithamam | aws1   | 99.90 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
```

DELETE

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

10 rows in set (0.00 sec)

mysql> DELETE FROM student_marks
      -> WHERE roll_no = 5;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 90.00 |
| 4       | Priya     | Math    | 88.90 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql> DELETE FROM student_marks
      -> WHERE marks < 75;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM student_marks;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 90.00 |
| 4       | Priya     | Math    | 88.90 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>
```

ORDER BY

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks  
-> ORDER BY marks ASC;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks  
-> ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks  
-> ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75

```
MySQL 8.0 Command Line Cli × + ▾

| 2 | Sita | Math | 92.75 |
| 3 | Anil | Math | 90.00 |
| 4 | Priya | Math | 88.90 |
| 1 | Ravi | Math | 85.50 |
| 10 | seetha | azure | 82.25 |
+-----+
9 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks
-> ORDER BY name ASC;
+-----+
| roll_no | name | subject | marks |
+-----+
| 3 | Anil | Math | 90.00 |
| 7 | dwaraka | dbms | 95.75 |
| 9 | kavithamam | aws1 | 99.90 |
| 4 | Priya | Math | 88.90 |
| 8 | ranjani | english | 97.40 |
| 1 | Ravi | Math | 85.50 |
| 10 | seetha | azure | 82.25 |
| 2 | Sita | Math | 92.75 |
| 6 | subbusir | aws | 98.50 |
+-----+
9 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks
-> ORDER BY name DESC;
+-----+
| roll_no | name | subject | marks |
+-----+
| 6 | subbusir | aws | 98.50 |
| 2 | Sita | Math | 92.75 |
| 10 | seetha | azure | 82.25 |
| 1 | Ravi | Math | 85.50 |
| 8 | ranjani | english | 97.40 |
| 4 | Priya | Math | 88.90 |
| 9 | kavithamam | aws1 | 99.90 |
| 7 | dwaraka | dbms | 95.75 |
| 3 | Anil | Math | 90.00 |
+-----+
9 rows in set (0.00 sec)

mysql>
```

GROUP BY

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

mysql> SELECT subject, SUM(marks)
-> FROM student_marks
-> GROUP BY subject;
+-----+-----+
| subject | SUM(marks) |
+-----+-----+
| Math    | 357.15     |
| aws     | 98.50      |
| dbms    | 95.75      |
| english | 97.40      |
| aws1    | 99.90      |
| azure   | 82.25      |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT subject, AVG(marks)
-> FROM student_marks
-> GROUP BY subject;
+-----+-----+
| subject | AVG(marks) |
+-----+-----+
| Math    | 89.287500  |
| aws     | 98.500000  |
| dbms    | 95.750000  |
| english | 97.400000  |
| aws1    | 99.900000  |
| azure   | 82.250000  |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT subject, COUNT(*)
-> FROM student_marks
-> GROUP BY subject;
+-----+-----+
| subject | COUNT(*) |
+-----+-----+
| Math    | 4        |
| aws     | 1        |
| dbms    | 1        |
| english | 1        |
| aws1    | 1        |
| azure   | 1        |
+-----+-----+
```

HAVING

```
+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, COUNT(*)
-> FROM student_marks
-> GROUP BY subject;
```

```
+-----+-----+
| subject | COUNT(*) |
+-----+-----+
| Math    |         4 |
| aws     |         1 |
| dbms    |         1 |
| english |         1 |
| aws1    |         1 |
| azure   |         1 |
+-----+-----+
```

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, AVG(marks)
-> FROM student_marks
-> GROUP BY subject
-> HAVING AVG(marks) > 90;
```

```
+-----+-----+
| subject | AVG(marks) |
+-----+-----+
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT subject, COUNT(*)
-> FROM student_marks
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
```

```
+-----+-----+
| subject | COUNT(*) |
+-----+-----+
| Math    |         4 |
+-----+-----+
```

```
1 row in set (0.00 sec)
```